

Multi-Department Budget & Asset Monitor

Comprehensive Independent Department Administration Manual

Document Scope: All Independent Departments	Target Audience: Department Admins & Leadership
Supported Depts: Finance, HR, Purchase, Quality, Sales, Ops, Prod, Logistics, IT	Database Engine: Supabase RLS PostgreSQL
Production URL: https://it-asset-phi.vercel.app	Brand Theme: HydraSpecma Gold (#FFCC00)

1. Multi-Department Architecture & Purpose

The **HydraSpecma India Budget & Asset Monitor** is designed to manage **every department independently** across the company. Rather than being limited to any single department, the platform provides dedicated budget management, asset registers, invoice records, and version tracking for **all 9 independent departments**:

- **Independent Departments Supported:** Finance, Human Resources (HR), Purchase & Procurement, Quality & Safety, Sales & Commercial, Operations, Production & Manufacturing, Logistics & Supply Chain, and Information Technology (IT).
- **100% Department Isolation & Ownership:** Each department operates with its own isolated annual budget figures, CapEx/OpEx allocations, asset records, invoice actuals, and revision histories.
- **Universal & Custom Categories:** Master expense categories (*e.g., Laptops, Workstations, Software Licenses, Machinery, Subscriptions, Services*) are available across all departments or customized per department.
- **Automated D365FO ERP Sync:** Connects fixed assets (`FixedAssetNumber` & `FixedAssetGroupId`) directly with Microsoft Dynamics 365 Finance & Operations.

2. Access Roles & Department Permission Matrix

The system enforces a strict **Role-Based Access Control (RBAC)** model backed by Supabase Row-Level Security (RLS). Users are assigned primary roles and cross-department permissions:

Role Name	Scope & Access Level	Key Capabilities
Global Admin	System-Wide Full Access	Manage all departments, configure user permission matrices, create/reset passwords, manage category masters, and access all budget data.
Department Admin (Dept Admin)	Full Access to Primary & Assigned Departments	Create & save budget figures, manage invoices, assign assets, run D365FO ERP syncs, execute bulk grid edits, and restore budget versions for authorized departments (*e.g., Finance Admin, Production Admin*).
Department Reader (Viewer)	Strict Read-Only Access	View budget vs actual figures, asset registers, and invoices for assigned departments. All edit buttons, save controls, and input boxes are strictly hidden/disabled.

Employee	Self-Service Personal View	Access personal assigned hardware/software items, warranty expiry status, and location tracking.
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3. Department Admin Operational Features

3.1 Independent Department Budgeting (/budgets)

The **Budget vs Actual** module allows Department Admins (*e.g. Production Admin, Finance Admin, Quality Admin*) to allocate annual budgets for their department across expense categories separated into **CapEx** (Capital Expenditure) and **OpEx** (Operating Expenditure).

- **Local vs. Global Scope Split:** Differentiates expenses for local department staff vs. global staff resources.
- **Real-Time Consumption Tracking:** Automatically aggregates actual invoice totals (`it\_invoices`) to compute consumed amounts and remaining balance.
- **Strict Read-Only Locking:** If a Department Admin switches to a department where they hold 'Viewer' access, budget fields convert into formatted read-only text and the 'Save budget' button is hidden.

3.2 Department Version Control & Revision History

Every time a Department Admin clicks '**Save budget**', the system automatically captures an immutable JSON snapshot of all budget lines (`v1.0`, `v2.0`, `v3.0`) isolated to that specific department.

- **Monetary & Remarks Diffing:** Compares selected versions against previous revisions to highlight exact line-by-line changes (e.g., *Local Budget*: ■5,00,000 → ■6,00,000 (+■1,00,000)).
- **One-Click Restoration:** Department Admins can select any historical snapshot and click '■ Restore Version' to revert active budget values instantly.

3.3 Department Asset Register & Microsoft D365FO ERP Sync (/assets)

The **Asset Register** tracks hardware, equipment, software, and peripheral inventory with employee assignments, department tags, purchase dates, warranty expiry, and budget allocation tags.

■ Microsoft Dynamics 365 FO (ERP) Interactive OData Sync

Department Admins can sync ERP Fixed Assets for their department seamlessly:

1. **Filtered OData Endpoint:** *FixedAssets?\$select=FixedAssetNumber,FixedAssetGroupId,Name,SerialNumber*
2. **Auto-Clipboard Detector:** Automatically reads copied OData payload from clipboard, parses records, and updates Supabase.
3. **Searchable D365 Asset Picker (`D365AssetPicker`):** When editing assets line-by-line or using **Bulk Grid Edit Mode**, typing keywords (e.g. 'Laptop' or 'HSINFA-000004') filters ERP assets in real-time to auto-fill Asset No & Group ID.

3.4 Department Invoices & Excel-Style Bulk Grid Editing

- **Invoice Tracking (/invoices):** Log PO numbers, invoice amounts, vendor details, and payment dates for your department.
- **Bulk Grid Edit Mode:** Enables simultaneous inline grid editing across multiple user names, emails, active statuses, asset numbers, and remarks with a single batch save.

- **Multi-Qty Asset Auto-Splitting:** Automatically splits bulk hardware line items (e.g., Qty = 10 items) into 10 separate 1-qty assets to assign to individual employees.

4. Security, Governance & Department Isolation

Data security and multi-tenant department isolation are strictly enforced across both front-end UI and back-end database layers:

1. **Supabase Row-Level Security (RLS):** All database queries (``it_budgets``, ``it_assets``, ``it_invoices``, ``it_budget_versions``) evaluate department policies.
2. **Dynamic Department Provider (``DeptProvider``):** Evaluates user permissions on department switch and sets ``isDeptAdmin`` state dynamically.
3. **Audit Logging:** Every budget save records creator metadata (``created_by``), timestamp, monetary delta, and version snapshot for compliance auditing.

5. Step-by-Step Department Admin Quick Start

Step	Action	Instructions
Step 1	Log In & Select Department	Log in at https://it-asset-phi.vercel.app/login . Select your target department (*e.g., Finance, Production, HR*) from the top bar dropdown menu.
Step 2	Update & Save Budget	Navigate to Budget vs Actual (/budgets). Input CapEx/OpEx category figures. Click Save budget to create a new Version snapshot (e.g., v2.0).
Step 3	Sync D365FO ERP Assets	Navigate to Asset Register (/assets). Click Sync D365 ERP -> Step 2: Open Filtered OData Feed . Press Ctrl+A, Ctrl+C. The system auto-detects and syncs database!
Step 4	Assign Assets & Bulk Edit	Click Bulk Grid Edit Mode . Use Search D365 ERP Asset... dropdown to select D365 asset numbers and assign staff members.
Step 5	Review Version Diffing	Click Version History to inspect revision diffs, line-by-line monetary variance, or restore previous budget states.